

Smart India Hackathon 2025 — Winning Solution

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MetaMark

AI-Driven Automated Legal Metrology Compliance Checker

Problem Statement: SIH25057 • **Ministry:** Consumer Affairs, Food & Public Distribution
Theme: Miscellaneous • Category: Software

This guide explains what MetaMark is, its features, and how to use it.

Contents

1. What is MetaMark?

2. Key Features

3. How It Works

4. Roles & Getting Started
5. Using the App — Consumer

6. Using the App — Seller

7. The Chrome Extension

8. Rewards & Meta-Tokens

9. Setup & Installation

10. Tech Stack & Architecture

11. FAQ & Troubleshooting

1. What is MetaMark?

MetaMark is an AI-driven compliance verification system that automatically validates **Legal Metrology declarations** on e-commerce products. It is built to help both consumers and sellers ensure that product listings correctly declare legal attributes such as MRP, net quantity, manufacturer details, country of origin, expiry, weight, and dimensions — in line with Indian Legal Metrology rules.

The platform combines **AI-based label analysis (Gemini multimodal vision)**, **Large Language Model (LLM) reasoning**, and an **automated compliance engine** to detect misleading packaging, incorrect declarations, and regulatory violations *before* products are published on a marketplace.

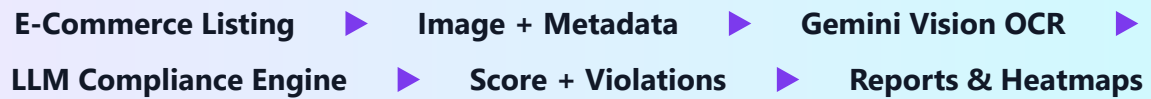
Who it helps

- **Consumers** — check whether a product they are buying is compliant.
- **Sellers** — verify a product is compliant before listing it, avoiding costly delisting.
- **Regulators / Platforms** — gain real-time, audit-ready compliance visibility and heatmaps.

2. Key Features

- **AI-Based Label & Declaration Analysis** — Multilingual OCR via Gemini Multimodal Vision extracts 25+ legal attributes from product images (MRP, net quantity, manufacturer, country of origin, expiry, batch, etc.).
- **LLM-Powered Compliance Reasoning** — Uses Gemini 2.5 Flash to interpret legal rules, reason about violations, and suggest remediation, adapting to updated regulations.
- **Confidence-Weighted Compliance Scoring** — Produces grades (A+, A, B, ...) and a detailed violation breakdown with an explanation-backed decision trail.
- **Pre-Upload Validation Engine** — Lets sellers validate listings before they go live, with readiness scores and fix suggestions.
- **Real-Time Compliance Heatmap** — Tracks compliance trends across manufacturers, importers, and packers; identifies repeat violators and high-risk entities.
- **Compliance Chatbot** — An interactive assistant that answers compliance questions with user/seller-context awareness.
- **Meta-Token Rewards** — Users earn Meta-Tokens (MT) for generating reports and purchases, redeemable for gift cards.
- **Chrome Extension** — A lightweight overlay that checks compliance directly on Amazon and Flipkart product pages.

3. How It Works



1. **Dynamic Web Crawler** — extracts live e-commerce listings and metadata (Amazon categories: general, food, skincare, electronics, books).
2. **Gemini Multimodal Vision OCR** — extracts textual and visual label information from product images.
3. **LLM Compliance Engine** — validates the extracted data against legal metrology rules.
4. **Auto-Remediation Engine** — maps each violation to an actionable fix.
5. **Scoring & Reporting** — generates explainable, audit-ready compliance reports.

4. Roles & Getting Started

MetaMark has two user roles, which determine what you can see and do after logging in:

Role	What they see
Consumer (customer)	Dashboard, Check Compliance, Entities (heatmap), Products, Chatbot, Rewards.
Seller	Seller Verification and Chatbot.

Sign up / Log in

1. Open the app (default `http://localhost:3000`).
2. Choose **Sign Up** and pick a **role** (*customer* or *seller*), a username, and a password.
3. After signing up, **Log In** with your credentials.
4. You are redirected to your role's home page. On login you may be given a `userId` and `role` ; the navigation menu adapts to your role.

5. Using the App — Consumer

Dashboard

Your landing page after login. Shows an overview of your recent activity, scraped products, and compliance status.

Check Compliance

Validates a product for legal metrology compliance. You can supply a product URL (or rely on scraped data) and trigger an AI analysis that returns a compliance **grade, score**, a list of high / medium / low priority issues, an estimated approval chance, and recommendations for fixing violations.

Entities

An interactive map/heatmap view that shows compliance trends across locations and entities, helping to spot high-risk sellers, importers, and packers. Powered by the backend heatmap APIs and Google Maps.

Products

Browse the products you (or your account) have scraped. View product details, detailed compliance analysis for a chosen product, and any stored product images.

Chatbot

An AI assistant tailored to your context. Ask compliance questions — it can answer general queries, explain issues on products you've checked, and give seller-specific guidance based on your role.

Rewards

See your accumulated **Meta-Tokens**, browse available gift cards, and redeem tokens for gifts. Your redemption history is shown here too.

6. Using the App — Seller

Seller Verification

Before uploading a product to a marketplace, sellers use this page to check that the listing will be compliant. It is a **fully software-based** process — no special hardware is required.

How to verify a product

1. **Choose a Product Category** — General (Amazon), Food & Beverages, Skincare & Cosmetics, Electronics & Electricals, or Books & Stationery.
2. **Enter a Product Description** — Describe the product in detail: brand, features, materials, usage.
3. **Enter Actual Weight and Dimensions** — Type the declared values directly, e.g. weight `250g` or `1.5kg`, and dimensions like `15x10x5 cm`.
4. **Add Product Images** — Either upload image files (JPEG/PNG, up to 10) or capture photos using your device's camera in the browser.
5. Click **Validate Product Compliance**.

What you get back

- Compliance **score** and **grade** (A/B/C, etc.).
- A summary of **high, medium, and low priority issues** with penalties.
- **Ready for Upload** (yes/no) and an **Estimated Approval Chance**.
- Actionable **recommendations** to fix any violations.

Software-only: The seller verification flow runs entirely in software — weight and dimensions are entered manually and images are uploaded or taken with a camera. It does not depend on external sensors or hardware.

Chatbot

Sellers can use the chatbot for compliance guidance and questions relevant to their listings.

7. The Chrome Extension

The MetaMark browser extension adds a compliance overlay to e-commerce product pages on **Amazon** and **Flipkart**.

What it does

- Detects product pages (e.g., `/dp/ASIN` on Amazon, `/p/...` on Flipkart).
- Injects a sticky MetaMark overlay and automatically runs a scrape + compliance validation.
- Shows a **grade / score badge** on the page and links to a full report in the web app.

Using the popup

- **Log in / Sign up** (with role selection) to connect your account.
- A **backend status indicator** shows whether the service is reachable.
- Toggle the extension **enabled / disabled**.
- A live **activity log** shows recent checks.

8. Rewards & Meta-Tokens

Meta-Tokens (MT) are MetaMark's native reward tokens that incentivize engagement.

Earning Meta-Tokens

- Generating a **compliance report**.
- Accessing **affiliate links** to partner e-commerce platforms.
- Successfully **purchasing products** via those links.

Tokens earned are proportional to the product value.

Redeeming

- Redeem tokens for **real-balance gift cards** on e-commerce platforms.
- Tokens are redeemable in fixed slabs: `10 MT`, `20 MT`, and `90 MT`.
- Gifts carry a `gift_code` and `gift_pin`; redemptions are tracked per user.

9. Setup & Installation

Prerequisites

- Python 3.8+ and `pip` (for the backend).
- MySQL 8.0+ (database name, e.g., `amazon_scraper_db`).
- Node.js / npm (for the Next.js frontend).
- A **Google Gemini API key** for live AI analysis (optional); without it the app runs in Demo Mode.

Backend (Flask)

1. Navigate to the backend directory: `legal_metrology_backend/legal_metrology`.
2. Install Python dependencies (e.g., `requirements.txt`).
3. Create a `.env` file with:

```
FLASK_SECRET_KEY=your_secret_key
DB_HOST=127.0.0.1
DB_PORT=3306
DB_USER=your_mysql_user
DB_PASSWORD=your_password
DB_NAME=amazon_scraper_db
GOOGLE_API_KEY=your_gemini_api_key
```

4. Initialize the database schema (`database_schema.sql` / `dump.sql`).
5. Run the server: `python server.py` — serves on `http://localhost:5000`.

API keys & Demo Mode

When `GOOGLE_API_KEY` is set, real Gemini analysis runs and `AI_AVAILABLE` is true. When it is absent, the app runs in **Demo Mode** and returns graceful placeholder compliance reports, which is useful for testing the interface without API costs.

Frontend (Next.js)

1. Navigate to the frontend directory: `frontend`.
2. Install dependencies with `npm install`.
3. Create a `.env.local` file:

```
NEXT_PUBLIC_BACKEND_URL=http://localhost:5000
```

```
NEXT_PUBLIC_GOOGLE_MAPS_API_KEY=your_google_maps_api_key
```

4. Run the dev server: `npm run dev` — serves on `http://localhost:3000`.
5. (Optional) Production: `npm run build` then `npm start`.

The frontend expects the backend at `http://localhost:5000` (via `NEXT_PUBLIC_BACKEND_URL`), and CORS is configured for `http://localhost:3000`.

Chrome Extension

1. Open `chrome://extensions`.
2. Enable **Developer mode**.
3. Click **Load unpacked** and select the `extension/` folder.
4. Ensure the backend is running so the extension can connect to `http://localhost:5000`.

10. Tech Stack & Architecture

Layer	Technologies
Frontend	React, Next.js (interactive dashboard)
Backend	Flask (Python), REST APIs
AI	Gemini multimodal OCR + compliance (google-generativeai / LangChain)
Web Crawling	Selenium, BeautifulSoup
Database	MySQL
Cloud	AWS / GCP, S3
Analytics	Python (statistical aggregation)

Key database tables include `users` (username, password, role), `products` (scraped listings + analysis results), `images` (product photos), `selleractivity` (scrape activity for heatmaps), and the reward `gifts` catalog.

11. FAQ & Troubleshooting

Reports say Demo Mode — what does this mean?

You have not set a `GOOGLE_API_KEY` in the backend `.env`. Add one to enable real AI analysis; otherwise placeholder reports are shown.

The frontend can't reach the backend

Confirm the backend is running on port `5000` and that `NEXT_PUBLIC_BACKEND_URL` points to it. Restart the frontend after changing env vars (they are read at startup).

Do I need sensors or hardware for the seller flow?

No. Seller verification is fully software-based: enter weight and dimensions manually and upload or capture images.

The extension can't connect

Ensure the backend is up and that the extension's host permissions cover the site you are browsing (Amazon/Flipkart). Check the backend status indicator in the popup.

Product scraping returns nothing

The scraper uses proxies/retries and headless-browser fallback. Slow or bot-walled sites may require `SCRAPER_*` tuning variables or retrying.